Uncle Jim's Sweet Factory

Lean 6 Sigma Yellow/Green Belt Course

Scenario & Exercises

Please don't take us literally!!

Scenarios are only simplifications and the one you are about to work with is no exception. Our aim is to allow you to practice some of the tools and techniques you will be learning on this course and demonstrate how they could be used in a real life setting. Unfortunately not every tool is used on every project which creates a problem as we wish to demonstrate the most important. Additionally, unless you know all the background knowledge that the project team would be aware of, the reason why a tool was used is often open to challenge. Don't waste time debating whether the scenario runs counter to your common sense or experience, just accept it as a tool to develop your skills and forgive us taking liberties from time to time.

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Uncle Jim's Sweet Factory

Consolidated Sugar & Fats (CS&F) is a manufacturer and supplier of bagged confectionary. The company has three major revenue streams:

1. **Uncle Jim's:** A high quality nostalgic product range. The company produces a number of "old fashioned" sweets which it bags and then sells to retail concerns world-wide. The company has established the brand as a luxury confectionary provider and uses only natural flavourings and ingredients in the recipe.

2. **Web Sales:** The Company also sells the "Uncle Jim's" brand direct to the public through their web site. Web sales are slowly growing however there is resistance to this new innovation partly because the culture of CS&F has always been one of tradition and limited change. Orders from the web are handled by the *Web Fulfilment Team* who take the order from the web site, pack and dispatch it.

3. **Target Sweets:** The Target Sweet range is a value range of "Wholesale Only" bagged sweets which CS&F sells in bulk to distributors (wholesalers). Target brand was devised to soak up the spare production capacity when the quotas for Uncle Jim's have been met. Target is "priced to sell" and basically duplicates the "Uncle Jim's" range but uses simpler and different packaging and synthesised flavours. Low cost is the main business driver for this brand. Uncle Jim's goes to great lengths to separate the two brands. The sweets are sold "informally" to a few UK wholesalers as and when a consignment is ready.

About the company

Uncle Jim's was founded in the 1920s and established a reputation as a manufacturer of quality confectionery, specialising in "boiled sweets" (sweets primarily composed of melted sugar and a small amount of flavourings).

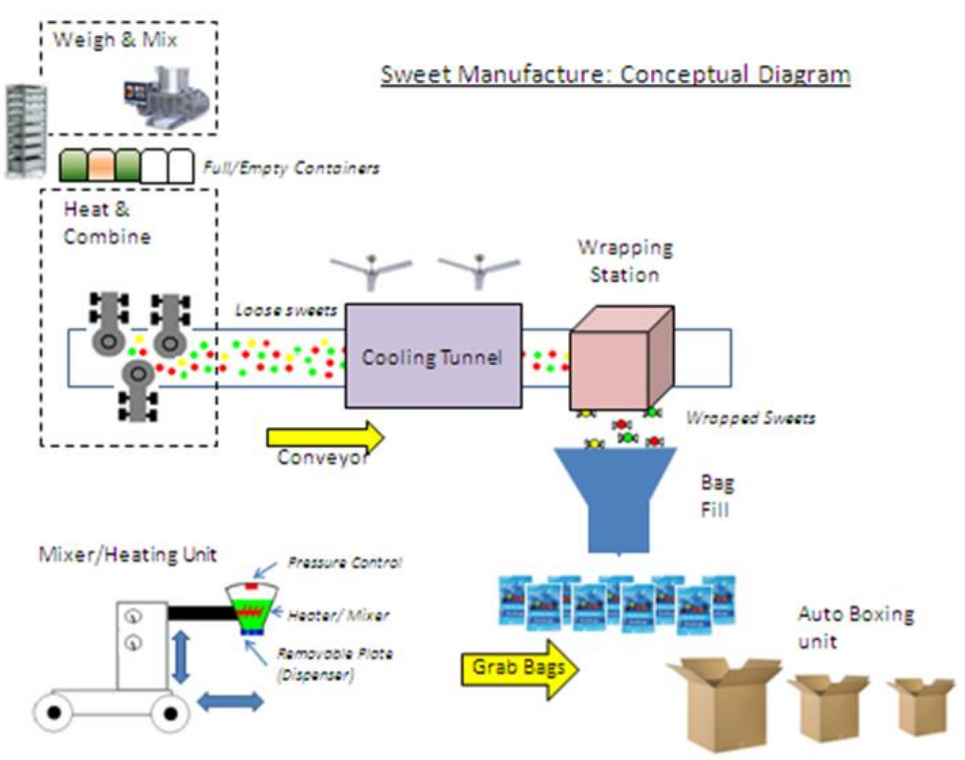
The original hand-made production process was replaced with two identical semi-automated production lines (described later) in the 1970s. The lines are comprised of units which are modular and robust and have remained in use up to the present time. The only significant change is the "Bagging station" – this is the module which decants the wrapped sweets into bags. This unit is much newer than the rest of the line and has more throughput capability.

While the manufacturing process is archaic by modern standards it does result in sweets with an "old fashioned" look to them. This is much prized by purchasers of the Uncle Jim's brand.

In the early 1990s the last family member running the company retired and sold the business to a consortium of senior managers. A limited company was formed - Consolidated Sugar & Fats (CS&F) although the trading name continued to be "Uncle Jim's", as did the brand.

A short overview of the sweet manufacturing process

Note: *This description refers to the manufacture of boiled sweets; the company has additional production lines for chocolate based sweets and candy, however these areas are not considered in this case study. Obviously the process has been simplified and shortened as the course is not about teaching you to make sweets!*



Each boiled sweet consists mainly of sugar with a small percentage of flavourings added.
Uncle Jim's produces and distributes their products from a central production facility. The raw materials for the sweets are obtained by the purchasing department and held in a central warehouse. The warehouse team keep a note of stock levels and when an item reaches a pre-determined figure they notify the purchasing department of the need to acquire another batch.

If the requirement is for sugar (the majority ingredient in most sweets) the purchasing department put the order out for open tender. They usually buy in bulk to get the best price. For the Uncle Jim's brand, flavourings are sourced from one specialist supplier with whom the company have a long standing relationship. Although flavourings make up a tiny proportion of each sweet, CS&F ensure that ethically sourced organic flavourings are used in their premium brand. Flavourings for Target sweets are obtained from the cheapest sources (even though the cost impact is minimal) and the cheapest quote is usually for synthetic flavours as natural/ethical products are up to 25% more expensive. To make sure that Target is a cheap as possible, the purchasing team have to expend a great deal of effort into tracking down and sourcing the lowest priced flavouring deals.

Ingredients are tested when they arrive at site to make sure they are of the required quality. If they are acceptable then they are stored until they are needed by manufacturing.

Manufacturing informs the warehouse every Friday of the ingredients needed for the following week. Over the weekend the requested quantities are assembled and shipped to the manufacturing holding bay early on Monday morning. The ingredients are once again tested (these tests only take about 30 minutes) and if still okay are used in manufacturing the sweets.
The production lines consume about 10-12 tons of sugar each day (purchasing order 200 tons as they get a discount for this quantity). The tender process for a new batch of sugar takes about 2 days from announcement to arrival at the site. Flavourings cycle between 15 and 45 days supply in the warehouse as the supplier for these has proved erratic in the past where their deliveries are concerned.
The sugar and the flavourings are melted in vats and the contents stirred. An automatic nozzle at the base of each vat opens for a timed period to allow a drop of the mixture to fall onto the rolling belt underneath. This is quite a slow process as it depends on the viscosity of the mixture so several vats are needed to sustain the flow. Each vat has an aperture control that can be raised or lowered; these adjustments set the size of the sweet and influence its shape. The mixture cools during its fall but is still soft enough to flatten on impact with the belt.

The production line has several vats so it is possible to create several different types of sweet at once; however the output of all the hoppers falls onto the same belt with the result that the sweets get mixed up. In practice this means that the production line is limited to **one** brand at a time however it is possible to produce several flavours at once if the intention is to bag the output as a mixture. The current practice is that if a mixture is to be bagged (e.g. Uncle Jim's Fruit Surprise) then the vats will hold different flavours, if only one type of sweet is needed (e.g. Uncle Jim's Old Fashioned Humbugs) then that mixture is spread across as many vats as required.

Once deposited, the sweets are cooled and passed to the wrapping station. The wrapped sweets are deposited into open containers which are manually moved to the Bagging Station. There, the sweets are transferred into a hopper which has a funnel at the bottom with an automatic dispensing valve. Opened bags travel under the valve which feeds 100 sweets into it. The bags then travel to a sealing station which closes the bags and deposits them into cartons of 500.

The factory has two production lines handling boiled sweets; this means that the company can produce two types of sweet at a time. To manufacture another type of sweet the factory needs to stop a production line and thoroughly clean it to remove any trace of the previous ingredients - this takes about 2 hours so company policy is to manufacture in large batches (typically they produce enough of one sweet to satisfy 3 months demand).

The filled cartons are transferred to the dispatch warehouse and are delivered by a variety of methods. When measured it was found that filled cartons could stay at the end of the production line for up to 4 hours before being transferred to dispatch.

Web demands are satisfied by a specialist team who act as a self-contained business within the parent company. They "purchase" cartons, split them and ship individual packets as required. A carton will last anything from half a day to a week depending on its popularity. The dispatch stock control system however treats the Web Fulfilment Team (WFT) as a wholesaler. Cartons cannot be picked up by the WFT direct but must be given to a third party transport company to deliver. The IT team have stated that they have several very important upgrades underway which have strict deadlines so they cannot deal with this relativity trivial matter for the next six months.

Report: The sweet production process @ 23rd April

(**Note:** *this section is only necessary for Exercise 11*)

The following is a report from an investigative team tasked with clarifying and analysing the current (AS-IS) state of the process flow.

The manufacturing process:

The sweets are produced as continuous batches during the day. Shifts are 05:00 -13:00 and 12:00 - 20:00.The shifts overlap in the middle of the day by 1 hour. This allows for an orderly handover and also for the early shift to tidy away and make good as necessary.

Production does not start until 06:00, although the first shift actually arrives at 05:00 each morning which allows the operators to organise themselves, make sure the first ingredients are ready and start up the line. A mechanic is also in attendance so that if the operators find any problems beyond their capability, a rapid fix can be assured. Production stops at 18:00 at which time the line is switched off and cleaned. Staff are paid two hours for this, but in practice the experienced teams usually only need one hour. Any serious faults are notified to maintenance who schedule repairs at the beginning of the morning shift (which sometimes starts late due to maintenance delays). During the day, as well as operations staff, there are two cleaners in attendance to keep the line clean.

Weigh & Mix

Batches are created in 250ltr lots which is the optimum size needed to fill a sweet boiler in the Heat & Combine stage. The weigh and mix team use 20 containers which they refill as required once the contents have been consumed by the next stage (Heat & combine). The weigh & mix team have three mixer units and each unit can mix up to 75 litres at a time. A cycle of weigh & mix takes roughly 6 mins per machine although this varies considerably with the experience of the operators.

Two operators are needed and depending on their experience can keep all three machines in continuous flow. The team discovered that the operators working in Weigh & Mix were more than able to satisfy the downstream demand. They noted that sometimes the operators were distracted from weighing & mixing by the need to collect empty containers from Heat & Combine (The H&C operators are supposed to stack the used containers at a central point but often “forget”).

Production have estimated that they waste about 5-15% of the ingredients at this stage over the course of a week (spillage occurs from time to time and mixed batches cannot be kept overnight so sometimes the last few containers of the day have to be thrown away). Different teams have their own way of working - some like to produce as much as they can at the start of a shift and then “coast” for the rest of the time, whilst others use a more planned approach. Often there are large amounts of mixture waiting to be collected by the Heat & Combine operators.

A sweet nominally weighs 5g, however there is considerable variation from operator to operator. According to the boiler operators this is inherent in their working practice and the crude machinery used (however this variance adds to the “old fashioned charm” of the product in CS&Fs customers’ eyes). The batches of sweet mixture are heavy so staff in both the Weigh & Mix and the Heat & Combine cells use mechanical assists to move and position the heavy containers. The team found that occasionally there was a shortage of the mechanical assists as they were not returned by users.

Heat & Combine

The heating units have limited capacity and take time to melt the sweet mixture. They also need to be fed constantly while in operation. There are three heating units which can be used.
Heat and combine is a continuous process as if a sugar boiler boils dry then the life of the heating coil is considerably shortened. At the beginning of the morning shift the boilers are filled and brought up to temperature. During the day they are topped up as required by their operators and at the end of the last shift they are left to cool down prior to being cleaned thoroughly.

The investigating team endeavoured to identify the wastage from the boilers. The best they could come up with was that for every 200 litres fed into a Boiling Unit (BU) about 160 litres of sweet mixture was dropped on the conveyor belt - the rest of this sticky material adheres to the unit itself and needs to be cleaned off. The three boilers are managed by a team of 2 operators.

The operators obtain the mixed material from the weighing station returning the empty cartons in the process. For a skilled team working with all three heating units; operating materials transfer takes about 15 mins each hour, the SMED for charging a single boiler takes 4 mins and cleaning & adjusting takes about 15 mins. The team works for a full shift staggering breaks as they wish. They also keep their area clean and perform basic maintenance in line with Lean Cell philosophy.

Thanks to a quality improvement initiative, it only takes the company 60 mins (of line downtime) to switch between brands – this is because the boilers need to be cleaned of their previous contents before refilling. At the end of the day the heating units need to be cleaned thoroughly (this takes 2 hours).

Cool & Wrap

The heated mixture drips onto the conveyor belt and flattens with impact. The combined output of all the heating units passes through a cooling tunnel for 4 minutes where filtered air is forced over them by fans. At the end of this process they pass into the wrapping station to be wrapped in plastic film.

The wrapping machinery is supposed to require 0.25 of a person to keep the plastic feed spools fed.

Unfortunately the machinery is considered temperamental and constantly jams so it has been found necessary to have a full time team person there to manage it. These units are the oldest machinery on the line. The Senior Mechanic considers that the installation was performed incorrectly by the manufacturer. He feels that the work was incompetent and sloppy and that the machines have never lived up to the manufacturer's claims.

The wrapping machinery seems to have been gradually getting more and more temperamental and is now failing to correctly wrap about 15% of the sweets as well as jamming frequently.
The cooled sweets are passed into the wrapping station where they are wrapped in plastic film. Hopefully a sweet is in and out of the wrapping station in about 15 seconds but unfortunately there seem to be problems with the machinery, leading to unpredictable delays.

On the day of the Gemba visit sweets could take anything from 15-45 secs because of various problems. The mechanics are at a loss to find a common cause for the problem, however one of the issues is that many sweets come out incorrectly wrapped. These sweets are treated as waste and thrown away. On the day the measurements were taken, approximately 15% of sweets were discarded in this way (the cooling tunnel is loss free).

Management is concerned that the time taken to cool & wrap is slowly growing as the wrapping station becomes increasingly troublesome. This is having an effect on the productivity of the whole line. The Senior Mechanic says "that's what you get for buying rubbish in the first place".
During the Gemba visit it was noted that due to mechanical problems the wrapping station was off-line for about 20% of each hour!

Bag & Box

Sweets from the wrapping station are poured into the collection hoppers. This needs to be monitored by staff as sometimes the hoppers overflow. The hopper's contents are transferred into the feed mechanisms which drop batches of approximately 100 sweets into bags. The bags are then sealed and deposited into cartons. The equipment used to do this is the newest on the line and is fully automatic. A team member is needed about 25% of the time to make sure that there are enough cartons in the feed mechanism. Occasionally bags are not correctly sealed (about 1 in every 500).

The company have a dedicated staff member operating this unit. As well as managing the hoppers the staff member identifies un-sealed bags and removes them before they are boxed. Faulty bags are opened and the contents placed back in the hopper.

Filled cartons (each carton contains 60 bags of sweets) are removed from the line and placed on pallets waiting for removal to the warehouse. This occupies one team member 50% of the time (one person handles the bagging stations of both production lines). This part of the production line is very efficient and has virtually no loss (0.00001%)

Dispatch

The filled cartons are transferred to the dispatch warehouse and are delivered by a variety of methods.

When measured it was found that filled cartons could stay at the end of the production line for up to 4 hours before being transferred to dispatch.

Typically cartons stay in the dispatch warehouse for up to 4 months before shipping.
There is some carton damage and about 0.5% of cartons are found to be split and not fit to ship.

Exercises

Exercise 1: How lean is my valley?

Read through the scenario section "**A short overview of the complete production process**" only, and identify potential areas where waste could be occurring.

Use the "Seven Wastes of Lean" slide as a guide.

Note: *This portion of the case study was included in your pre-course reading so you should be familiar with it.*

Exercise 2: Identifying the VOC

Note: *This is quite a long exercise with a lot of instructions. We suggest reading the whole thing through before you start, then discuss any points that need clarification with your Instructor.*

Consider the following three Customer Archetypes

Mr Brown

Is a middle aged man who has been married for 15 years. He and his partner have 2 small children and after essential outgoings have negligible disposable income. Occasionally the Browns get a baby sitter and go to the local picture show, and to mark these occasions usually take a pack of sweets with them. Mr Brown also likes to buy sweets occasionally for his partner and for his elderly Mum. He is a nice man who takes a lot of pleasure in giving those he loves nice surprises. He is the sort of chap who if he won the lottery would give most of it away to his friends & family.

Hotel Swish

Is a small boutique style hotel. Whilst they break-even during the week, they make most of their profit at weekends by offering romantic getaway breaks to couples. They are also licensed to perform weddings. The Hotel provides sweets in their hotel rooms and there is also a bowl on the receptionist's desk.

Jones Corner Shop

Jones keeps open long hours and serves the local community. As well as providing staples to the district they also stock a few luxury items. Jones cannot compete on price with the big supermarket 3 miles away, so it has to focus instead on convenience and quality.

Activity 1:

Consider the list of comments made by the three Archetypes as shown in the table following.

Use the comments, the descriptions and your common sense to identify features which each archetype would want in a bag/box of sweets. Write each feature on an individual sticky note and make sure the customer is also identified. You will often find that several archetypes will want the same feature (or something very similar) in which case create a sticky for each archetype who wants that feature. Don't worry about having several features that are the same (or very similar) from different archetypes. This is to be expected and we would also expect that there would be a large overlap between different customers (everyone wants a pure milk don't they). Put in anything and everything, from essential features up to the most wishful ones. If you can think of any additional features feel free to add those as well.

So in summary you should end up with three piles of stickies, one pile per customer with each sticky describing one feature that that customer would like the product to have. You will have lots of similar features in the piles.

Comments		
Mr Brown	**Hotel Swish**	**Jones**
I would rather have smaller good quality sweets rather than lots of cheaper bigger bags	We need the ability to provide small packets of sweets to our guests as bedroom favours	We need to appeal to impulse buyers so we want attractive packaging that is easy to place by the sales tills
Both my Mum and my partner like the fancy packaging	It would be nice if we could get big bags of the sweets to top up the bowls in reception	Small treats that are easy to pick up would be ideal
Money is tight and I would rather buy little and often as a treat if I can	The impression of luxury is important to us as it helps build the image of our hotel in customers eyes	Gifts and promotions such as “2 for 1s” always go well with the customers
A £2 packet should contain at least 100 sweets although I would expect to find 150 in them	Our customers are very keen on sustainability and ethical “fair trade”	It would be nice if the packaging would promote the product. A real bonus is if the packaging can be converted easily into a presentation display
Modern sweets are all identical shapes – that doesn’t look “old fashioned” to me! Still I suppose that making them the old way would be too expensive	We need packs that sit nicely on the pillows, are easy to store and are vermin resistant	It’s amazing how many packs of sweets you can shift due to “pester power” – we place them by the till and if the small (pre-school) kids notice them they can usually get their parents to buy a pack
Treats should be gift-wrapped	It would be wonderful if we could have our details printed on the pack (at no extra charge)	The sweets will be left on the counter overnight so I don’t want them to be contaminated by mice
Really nice to get a surprise or a gift in them – I remember when we won a weekend away in a pack of crisps	I do wish the shelf life was longer – I had a customer the other day who complained that the sweets had only 3 months left before their end of life!	Must be under £2:50 but it would be best if less than £2 (for impulse buyers). If I could get a reasonable pack at £1 retail then they would fly off the shelves.

Take the stickies you have just created and use them to build a Customer/Feature Diagram (instructions to create a Customer/Feature Diagram are included in a FYI slide in the notes).

Mark out the zones (Basic, Expected, Delighter) and by consensus move the features into each zone.

This is where you will begin to see differentiation between customers as you identify which features the archetypes have in common. Note that even if two customers want the same feature their priorities may be different.

Activity 2:

Using the Customer/Feature Diagram created in Activity 2, create an Affinity Diagram (instructions to create an Affinity Diagram are included in a FYI slide in the notes). Look to identify and harmonise common themes as your clusters.

Using yet more stickies, provide group descriptions that harmonise the collections of features in each group.

By consensus identify if the group descriptions should be Basic, Expected, or Exciters. Remember these will be global so if one customer has identified a feature as a basic then globally you should set it as basic.

Activity 3:

Using the following memo from CS&F Marketing and the outputs from Activities 1 & 2, create a Kano diagram showing how you think each theme should be represented from the customer's point of view. Use the group stickies from Activity 2 and estimate each stickies position on the appropriate tram line.

Alternatively you can use a Feature/Summary diagram.

Uncle Jims product lines (Memo from CS&F Marketing)

At the moment there are 15 different flavours and mixes of sweets produced under the "Uncle Jim's" brand. All sweets are packaged in plastic packs ("grab bags") as these are what our Supermarket Customers prefer. Bags hold 100 sweets on average. All ingredients are ethically produced from sustainable sources and are 100% organic. We are certified by reputable organisations for the above and have permission to use their accreditation Logos on in our marketing. While we do use the accreditation logos on our web site we do not have them on our Grab Bags. The cost would not be significant to put the logos on although it would need some effort to redesign the artwork however. Speaking of artwork, the images, colours and fonts etc are designed to appeal to the mature customer who is prepared to spend a little more to get something similar to what they remembered from their youth. We do have the capacity to put out sweets into small sealed boxes. The cost of switching to a box from a plastic bag adds about 20p to each product. We can only do small and medium quantities in boxes however. We can also do large capacity old fashioned jars. We used to sell these to old fashioned sweet shops, however we discontinued this as it was very expensive and prone to transit damage.

Exercise 3: Drafting a SIPOC

CS&F's purchasing department is responsible for acquiring the ingredients for sweet production. Their process for ordering a new batch of sugar is as follows:

The warehouse notifies purchasing by e-mail when stock has fallen below a pre-set limit and this starts the process. The e-mail is picked up and the task of generating an order is allocated to whoever has the time.

CS&F are members of an on-line B2B (Business to Business) web portal. The purchasing team member logs on and places a tender (an invitation to quote). A quantity of 250 tons of sugar is requested and the properties of the sugar specified. CS&F use a specific type of sugar (as defined by the CS&F manufacturing team). Also the purity of the sugar used has to conform to current government regulations as defined by the Food Standards Agency (FSA). The relevant details are included in the tender. The tender document also includes the required delivery date.

2-3 days after the tender has been placed it is closed and the lowest bid accepted. A note is added to the finance system identifying the shipment but flagging it as "do not pay invoice without clearance from purchasing"

Once the sugar is delivered to the warehouse, it is tested and if satisfactory, purchasing are informed.

The shipment record on the finance system is now changed to "pay on invoice".

If the sugar is not tested as satisfactory, purchasing contact the supplier and request a replacement shipment.

Your task

Using the scenario and the outputs from former exercises construct a SIPOC for the above process

Alternatively create a SIPOC for a process you are involved with at work

You can draw up your SIPOC on the next page

You can draw your SIPOC here

Exercise 4: Making light of Risk

The Finance Business Unit of CS&F is based on the top floor of the main office block. This is a four story building with a fragile flat roof. The building itself was built in the 1950s and has a large number of big glass windows, which usually provide a good level of natural light for the staff working there. However, if a cloud obscures the sun the illumination inside the building changes noticeably. In actual fact there is usually adequate lighting even if the day is cloudy - it is just that the sudden difference is very noticeable. When a change occurs, staff have got into the habit of switching on the lights, but then if the clouds then dissipate they often forget to switch them off again. This is a waste which the company is keen to remove. An improvement plan has been suggested as follows:

1. All manual light switches to be removed
2. In future the lights will be controlled by a timer and a light sensor installed on the roof
3. The sensor will automatically switch on the lights when the outside light falls below a predetermined level and switch off when the light level rises
4. The timer will keep the light switched off outside working hours (all weekends and Mon-Fri between 18:00 and 07:00 hours, regardless of the sensor). The timer acts as a master, switching on or off all lights in the Business Unit and is a simple mechanical system situated in the main office (illustration on the right)

The factory is located in a semi-rural area and on windy days quite a lot of dust is blown up from the countryside

The roof is fragile, so to fix the sensor the electricians will need to place temporary crawl boards on the roof. The nearest source of power for the sensor is a large Illuminated Sign advertising Uncle Jim's Brand also mounted on the roof.

(Picture of the roof from the sign is shown on the right)

Your tasks

1. Identify two risks which could have a negative effect on CS&F from this implementation project (you can make your notes on the following page)
2. For each risk identify:
   a) An EWI (Early Warning Indicator)
   b) A method of mitigation (identifying any secondary risks resulting from your proposed action)
   c) An assessment of any remaining residual risk

Present your conclusions in the class discussion which follows this exercise.

Exercise 4 Notes:

Exercise 5: Collecting Data

Management are concerned about the high volume of wastage arising from the boiled sweet production process.
An improvement team has been formed and tasked with improving the manufacturing process. The Team has identified several areas which they think might contain waste, however they need to collect more information so that they can “manage from fact”.

The Team has been given clearance to move to the Measure phase and needs to examine the **AS-IS** state and create a Value Stream Map.

The main areas that the team want to investigate are:

- High number of incorrectly wrapped sweets
- Mis-shapes (sweets incorrectly formed)
- Incorrect bag fill

Your Tasks

Create a Data Collection Plan and identify the following:

a) What data will need to be collected?
b) How it will be collected?
c) Over what period would you recommend collecting the data?
d) What sort of sample size would you recommend?
e) How often would you recommend sampling?

For c, d and e, use your common sense and attempt to balance the cost of collecting data against your need to obtain enough data to model the process sufficiently so that you can understand it properly.

Note: *Do not worry about identifying the Data Type for this exercise. We will cover this topic later in the course.*

Some rules of thumb to help you:

Look for blocks and boundaries that might illuminate the problems you are investigating, for example, are shifts or days significant?

Can you identify any cycles in the process (ask the people in the process or investigate what customers think)

Attempt to sample at least 4 four times in a cycle or block and measure over 4 cycles if you have the time

You can draw up your Data Collection Plan on the next page …

You can draw your Data Collection Plan here

Exercise 6: Production Problems

Note: *Your instructor will assign you to a team for this exercise. As a team you should consider splitting up the work between yourselves so that one group performs the calculations for Production Line A, the other for B, then you come together to contrast & compare*

There is concern over the variation in the number of sweets placed in each bag by the automatic fill & seal system

During the day 30 samples are taken at regular intervals from both production lines using the following data collection method: one bag is removed from each assembly line and the contents checked. The results from the two production lines have been collated in the following table:

Production Line A							**Production Line B**					
101	103	102	103	115	102		99	101	96	98	99	100
101	103	103	103	99	102		95	99	103	102	101	100
100	101	100	105	103	102		99	98	97	96	98	102
99	104	102	104	99	100		99	101	101	100	99	102
104	103	98	102	103	106		100	99	100	98	101	97

Voice of the Customer

Current Trading Standards law mandates that sweet bags showing a nominal content of 100 must contain at least 98 sweets

Management want to restrict overfill to a maximum of 2 sweets in order to contain costs

Activity 1

The team working on this project have taken the raw data, calculated the mean and produced a Frequency Table for each production line.

Using the Frequency Table for the appropriate production line, create Histograms showing the distribution of each line.

Production Line A			**Production Line B**	
Bag Contents	**Frequency**		**Bag Contents**	**Frequency**
			95	1
			96	2
			97	2
98	1		98	4
99	3		99	7
100	3		100	5
101	3		101	5
102	6		102	3
103	8		103	1
104	3			
105	1			
106	1			
107-114	0			
115	1			

You will need to calculate the segment size (use the “rule of 10” described in the course plus your common sense).

Contrast the two lines together and answer the following questions

1. Do you think that the data collected follows the normal distribution curve?
2. What is the range (minimum and maximum fill)?
3. From (2) what is the spread of variation (difference between maximum and minimum)?
4. Give an opinion on Process Capability (how much does VOP meet the VOC?)
5. Can you tell if there are any **special causes?**
6. The Mean for Production Line A has been calculated as 102 sweets while that for Production Line B is 98. Do you think that the Histograms support these calculations?
7. Do you think the two production lines have similar output variability?

Notes

Activity 2

The next step is to Chart the process. The Means for each production line has been calculated and the SD has been estimated from the range.

Create a control chart for each line. Use +/- 3σ as your control limits.
Use the following figures for your chart

- For both production lines use a SD of **two sweets**.
- Mean (BAR-X) for Production Line A = 102 sweets
- Mean (BAR-X) for Production Line B = 99 sweets

Sketch in the USL and LSL on your chart

Answer the following for both production lines:

1. Can you identify any special causes?
2. How well does the production meet the customers' requirements?

TIPS: How to create the control chart

1. Use the table showing the 30 individual plots for your production line (NOT the frequency table)
2. You can use your own graph paper or the graph paper on the following page
3. Work out your scaling – where on the paper you will place the mean average line, is there room for the UCL & LCL, how far apart horizontally will each of you 30 plots be?
4. Draw in the mean average line
5. Using an estimated SD of 2 (sweets) mark in +/-1SD, +/-2SD and +/-3SD lines
6. Check that everything fits
7. Start marking in your data points. The order the data collected is important so read across the tables (L➔R) starting at the top row
8. Take the first value (101 in Production Line A) and pace this on your grid where point one on the horizontal axis intersects the 101 value on the vertical axis
9. Do this for the next 29 plots
10. Draw a connecting line between the plots
11. Look for potential special causes using the JQT and the SD lines as a guide

Activity 3

As part of their quality checking the Compliance team monitored the defects on Production Line B.

The team took a sample of 1000 sweets and assessed them against the criteria of:

- Wholly wrapped
- Acceptable shape
- Correct weight

They found that a total of 8 sweets were outside requirements as follows:

Sweet	Partially wrapped	Unacceptable shape	Too heavy	Too light
1	X	X		
2		X		
3			X	
4			X	
5		X		X
6			X	
7	X			
8	X	X		

Using the above information as required, calculate the DPMO and the Long Term Sigma level of this process

Activity 4

The Quality Improvement Team is considering two hypotheses for possible root cause(s) of the variation in bag fill. The two hypotheses which have been identified are:

1. Changes in the speed of the production line are causing the variation in bag fill

2. Changes in the weight of the sweets is causing a variation in bag fill

A data collection plan is prepared and the following samples taken:

SPEED & FILL			
Speed (mm/S)	**Fill**	**Speed (mm/S)**	**Fill**
10	101	12	103
12	97	12	103
13	100	13	101
13	99	13	99
10	104	13	103
10	98	14	99
10	101	14	100
11	96	14	98
11	100	10	99
12	100	11	101

mm/S = Millimeters per Second)

Note: *Read **Down** the columns*

Data Collection Plan

20 bags sampled at random and the speed of the production line identified. Contents then counted.

WEIGHT & FILL			
Weight	**Fill**	**Weight**	**Fill**
4.87	101	4.65	104
4.63	101	4.79	103
4.97	100	5.39	97
5.10	99	4.71	103
4.86	102	4.58	105
5.04	99	4.86	101
5.02	100	5.20	98
5.20	96	4.99	100
4.91	101	4.99	99
4.89	100	4.72	103

Data Collection Plan

20 bags sampled at random

Contents counted and mean weight of the sweets calculated in grams to 2 decimal places.

Nominal weight of the sweet is 5g

Using the above data construct a Scatter Diagram and investigate any possible correlation between the speed of the line and the fill, and the weight of the contents and the fill (you will need to create two scatter diagrams)

You can use the following two sets of graph paper if you wish

Exercise 7: Identifying the Value Stream

Using the case study and the additional section titled “Report: The sweet production process @ 23rd April”, construct a Value Stream map of the whole process showing the “As Is” (current) state. Use the blocks as outlined in the report as the basis for your process map and, using the information included in the report and your common sense, see if you can identify:

- potential areas of waste
- The most likely bottleneck
- Which process block you would look to improve first

The basic flow is similar to the Headache tablet VSM discussed in the course.

Do not attempt to go down into too much detail, for this exercise it is sufficient to stick to the basic blocks outlined in “Short overview of the complete production process” section of the Case Study

For the purpose of this exercise assume that the production line made only one brand of sweet that day

The Case Study contains some useful information about the process that will allow you to identify some areas of potential waste.

Exam tips & Hints

General points

The aim of our exams is to allow you to showcase your knowledge of Lean 6 Sigma. We do not want to waste our time or your time in the classroom by having to explain the format of exam questions or how the exam works. We have therefore opted instead for a simple exam framework, so that if you know the subject material, then you should be able to answer the questions.

Both exams are open book - that is, you can refer to the course workbook, your notes and anything else provided by your instructor. We call this the “BOK” (Body Of Knowledge) and when we write the exam questions, this is what we base them on. Please be aware that Lean 6 Sigma has never been rigidly defined and there are often many different interpretations and/or definitions of many topics and their associated tools/techniques. So if you use knowledge gained prior to or outside of the course, or additional material, then it is possible that you may be led astray. For this reason, we only allow materials provided during the class to be used during the exam.

Yellow Belt

Each question is worth 2 marks and has only **one** correct answer from four provided. You will be tested on straightforward facts and will also be expected to identify the tools described in the BOK from a description.

Green Belt

We test for a greater level of understanding in the GB exam. The syllabus for this exam includes the Yellow Belt BOK, so if you attended a YB somewhere else, please do make sure that you are aware of the contents of **our** courseware.

The Green Belt exam consists of a mixture of multiple choice questions (similar to the Yellow Belt but with five possible answers instead of four) and some additional opinion based questions requiring a **short** written answer. All multiple choice questions only have one answer although sometimes you will have to scrutinise the answer set carefully. With the written questions there is usually more than one correct answer and a lot will hang on the explanations and reasons given to justify your answers.

While the exam is open book we will be disappointed if you just copy straight out of the workbook. Have a look at our examples for guidance.

Tips for the exam

Read and understand the question and answer the question

It is important to understand what you are being asked – without this you cannot select the correct answer or write a suitable response.

In opinion based questions make sure you give the correct information

We constantly have to mark down answers that do not give us the information we have asked for. This is frustrating, as we usually know that the candidate has the knowledge but they just forgot to put it down.

Show your working out

This means if you perform a calculation incorrectly we can track down were you went wrong and award marks for what you did right.

Examples

Q (2 marks)

According to the BOK which statement is true about Lean 6 Sigma?

- **A.** Application will remove all waste from a process
- **B.** Costs will be dramatically reduced
- **C.** Yellow Belts remove road blocks
- **D.** Lean came from TPS

D is the correct answer. A would be incorrect as we will not be able to remove "all" waste. B is very specific and cannot be justified from the BOK or the question – it may be that costs are reduced dramatically, or it may be that we increase our costs as that is what the customer would prefer, e.g. do you want really want your holiday plane to be serviced by the cheapest people? C is technically incorrect as it is Champions who remove roadblocks, not YB's.

Q (4 marks):

Identify two Lean tools and provide an explanation (max two bullet points) for each one.

Bad Answer 1:

SIPOC
Poka Yoke

Why this is bad: *There is no explanation of the tools which is the most important part of the test. The best you could hope for here is 1 mark.*

Bad Answer 2:

Control Chart – Models a process using a time dimension
Pie Chart – Shows the percentage count between boundaries

Why this is bad: *These tools are from the Six Sigma part of Lean 6 Sigma. Do check whether the question is asking specifically for Lean Tools, Six Sigma Tools or Lean Six Sigma Tools*

Q (3 marks):

Explain the function of a SIPOC.

Bad Answer
SIPOC – Suppliers, Inputs, Process, Outputs, Customer

Why this is bad: This does not *explain* the function it just describes its components. You will pick up 1 mark at best.

Good Answer
A SIPOC is a high level overview of a process or part of it identifying the relevant Suppliers, Inputs, Outputs and Customers and a high level process map. It is used as a starting point for the team to begin understanding what they will be improving and make sure they have not taken on too much.

Q (5 marks):
Would you use a HPT in your workplace? Briefly justify your answer.

Bad Answer 1

Yes I would

Why this is bad: *No justification*

Bad Answer 2
A HPT requires a great deal of effort to set up. It consists of up to 9 people working closely together over a medium-long period of time. Relationships are strong and enduring and will develop as the team progresses. They use a self-managed model for group direction, usually based around daily Stand-ups and perhaps Kanban boards.

Why this is bad: We do not know whether the candidate is for or against here. We have some really detailed explanations which correctly describe how HPT's may work, but which don't actually answer the question, and give us no sense of whether the writer would use HPT's in their workplace and why/why not. At best we can give perhaps 2 marks when we know we should have been able to give 5.

Q (5 marks):
Give an example of ***where*** Poka Yoke can be used in the CS&F production line.

Bad Answer (1)
Poka Yoke is mistake proofing – preventing errors from occurring. These should be low cost and physically prevent errors if possible.

Why this is bad: *This is describing the technique, NOT how or where it can be applied.*

Bad Answer (2)
CS&F could use Poka Yoke to prevent mistakes happening on the production line.

Why this is bad: As examiners we cannot tell if you can actually identify ***where*** this technique can be useful.

Good Answer
In my business (Catering) we use high visibility (blue) plasters on cuts not flesh coloured. This means we are alerted when one falls off.

Why this is good: It demonstrates the candidate can use this technique when they go back to work. Note that in this case, the candidate chooses not to give an example from CS&F as they felt more at home with their workplace. As examiners, we will accept this alternative, as all we want is for you to demonstrate your understanding of the topic.